# Thomas Crowe

## Full-Stack Developer & Technical Operations Lead

Full-Stack Development • React • Node/Express • PostgreSQL • PHP • Python

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## SUMMARY

Full-stack developer who designs, ships, and operates production software for a B2B e-commerce manufacturer — internal React/Express/PostgreSQL platforms, a custom WooCommerce storefront, and integrations with accounting, shipping, and manufacturing systems. I work across every layer, from UI to database to deploy pipeline, and my automation has eliminated ~450 hours of manual document work.

## EXPERIENCE

**Full-Stack Developer & Technical Operations Lead** | CalibrationWands.com (Fastec Services LLC) Feb 2024 – Present

*Travelers Rest, SC*

Build and operate the technical stack for a B2B manufacturer of NIST-traceable test standards serving food, pharmaceutical, and textile quality-control customers.

- **Test Standards Hub** — built an internal full-stack platform (React/Vite, Express, Prisma/PostgreSQL) that turns WooCommerce orders into manufacturing outputs: 3,800+ auto-generated NIST-traceable certificates (~450 hours of manual document work eliminated), engraving SVGs, and STL files sent to a CNC machine. Certificates are generated through Adobe PDF Services, AES-256 locked, and cryptographically signed (pyHanko) for tamper-evident verification.
- **WooCommerce platform engineering** — 50+ custom plugins: Stripe checkout guards, pay-on-PO terms with credit limits, quote-to-order, certificate portal, schema.org SEO; feature-flagged deploys with automatic rollback. Ported the SKU generator to PHP with byte-identical output across all 7,093 product configurations.
- **Custom storefront engineering** — dynamic SKU-configurator across 10 product lines (JavaScript + PHP WooCommerce hooks) generating valid SKUs in real time, with cart, order-meta, email, and PDF integration; debugged a cache-induced add-to-cart race condition; Dompdf invoice templates, Stripe payments, Playwright end-to-end tests.
- **Accounting & inventory integration** — connected the order platform to QuickBooks Desktop (Web Connector/qbXML) for invoices, estimates, items, customers, and inventory adjustments, with per-write verification, an audit change log with revert, and a provider-agnostic accounting interface built for a future system migration; integrated Zoho Inventory so stock is decremented and builds are planned automatically as certificates are generated.
- **Order-to-manufacturing automation** — WordPress webhook over a Cloudflare Tunnel triggers certificate generation and PrintNode shop-floor printing on every order, generates UPS labels (including international export docs) with tracking written back, and serves finished certificates through an account-gated customer portal — no exposed origin and no manual hand-off.
- **Zero-downtime production migration** — migrated a live B2B store to dedicated DigitalOcean NVMe hosting with no downtime or data loss, cutting recurring outages from 145+/month to near zero (Cloudflare DNS/CDN, Redis Object Cache Pro, restored SPF/DKIM/DMARC), then added a Git-deployed Bedrock staging environment for safe pre-production testing.

## PROJECTS

- **FlowLocal** — Open-source local voice-dictation stack for Windows — GPU faster-whisper transcription, Ollama text cleanup, retrieval over personal notes, and streamed Kokoro TTS replies (FastAPI), fully local. github.com/wtcrowe4/FlowLocal
- **Agent Workstation** — Daily-driver tooling for directing AI coding agents: DialDeck Surface Dial control surface for herdr agent orchestration (AutoHotkey v2), a local Kokoro TTS daemon (FastAPI), and a herdr message-inbox plugin. github.com/wtcrowe4/DialDeck

## SKILLS

**Languages:** JavaScript/TypeScript, Python, PHP, SQL, Bash, PowerShell, C#, HTML/CSS
**Web:** React, Next.js, Node.js/Express, REST APIs, Three.js/React Three Fiber, Tailwind, Vite
**Data:** PostgreSQL, MySQL/MariaDB, SQL Server, MongoDB, Prisma, pgvector, Python ETL, FastAPI
**Testing & Quality:** Vitest, Node test runner, Supertest, Playwright (E2E), ESLint, Git code review, CI test gates
**Infrastructure & DevOps:** Docker, Linux/systemd, GitLab CI, GitHub, Cloudflare (DNS/CDN/Tunnel/Access), AWS, DigitalOcean, Oracle Cloud, Nginx, Sentry, Tailscale, SOPS/age
**Business Systems:** WordPress/WooCommerce, Stripe, QuickBooks Desktop (qbXML), Zoho Inventory, UPS API, PrintNode, Google Workspace
**AI & Automation:** vLLM, Ollama, LangGraph/LangChain, RAG (pgvector), PyTorch, Unsloth fine-tuning, AWQ/GPTQ, Langfuse, Whisper, MCP, Claude Code agents, CodeRabbit, pyHanko

## ADDITIONAL EXPERIENCE

**General Manager**, Arnold's of Powdersville — Powdersville, SC 2016 – 2024

Daily operations, budgeting, inventory, and team leadership for a high-volume restaurant; Clover POS administration and troubleshooting.

## EDUCATION

**North Carolina State University** — Software Development Program, Computer Software Engineering 2022
**Clemson University** — Computer Engineering 2010 – 2013
**College of Charleston** — Honors College, Biochemistry 2007 – 2010